1. S. Koenig, *Sociology: an Introduction to the Science of Society*. Barnes & Noble 1957

2. I. Robertson, *Society: A Brief Introduction*, 1st ed. Worth Publishers, 1988

Math 4141 Geometry and Differential Calculus Credit: 4.00

2D Co-ordinate Geometry: Change of axes: transformation of coordinates. Simplification of equations of the curves. Pair of straight lines: Homogeneous second degree equations. Conditions for general second degree equations to represent a pair of straight lines. Angle between the lines. Pair of straight lines joining the origin to the points of intersection of the curve and a line. Circles and system of circles: Tangents and normals. Pair of tangents. Chord of contact. Orthogonal circles. Radical axis and its properties. Parametric coordinates.

3D Co-ordinate Geometry: Rectangular coordinates. Direction cosines and angle between two lines. The plane and the straight lines. The equation of a sphere. The standard forms of equations of the central conicoids, cones and cylinders.

Differential Calculus: Limits, Continuity and Differentiability. Differentiation of explicit and implicit function and parametric equations. Significance of derivatives. Differentials, Successive differentiation of various types of functions. Leibnitz's theorem. Rolle's theorem, Mean value theorems. Taylor's theorem in finite and infinite forms. Maclaurin's theorem in finite and infinite forms. Langrange's form of remainders. Cauehy's form of remainder. Expansion of functions by differentiation and integration. Partial differentiation. Euler's theorem. Tangent, maximum and minimum values of functions and points of inflection. Applications of Differential Calculus. Evaluation of indeterminate forms by L'Hospitals rule, Curvature, center of curvature and chord of curvature. Evolutes and involutes. Asymptotes. Envelopes. Curve tracing.

Recommended Texts:

1. H. Anton and A. Herr, *Multivariable calculus*, 5th ed. Wiley, 1995

2. S. L. Loney, *The Elements of Coordinate Geometry*, 11th ed. Macmillan and Company, 1908

3. E. W. Swokowski, *Calculus with Analytic Geometry*, 4th ed. Prindle, Weber & Schmidt, 1988

Phy 4143 Physics II Credit: 3.00

Electrical Units and Standards. Electrical Networks, circuit solutionsseries, series parallel networks, loop and Nodal methods. Delta-wye Transformation, Circuit Theorems: Superposition theorem, Thevenen's and Norton's Theorem. Concept of Dual Networks.

Basic principle of generation of Alternating and Direct Current, Introduction to phasor algebra as applied to A.C. circuit analysis. Solution of A.C. circuits: Series, Parallel and Series-Parallel circuit, R.L.C circuits series and parallel resonance. Applications of Networks theorems to A.C. circuits.

The magnetic intensity, flux/density, magnetic effects of Electric current, Magnetic circuit concepts, BH curves, characteristics of magnetic materials, magnetic force and its utilisation, Hystersis and eddy current losses, magnetic circuit with A.C. and D.C. excitation.

Recommended Texts:

1. C. K. Alexander and M. N. O. Sadiku, *Fundamentals of Electric Circuits*, 6th ed. McGraw Hill, 2016

2. R. L. Boylestad, *Introductory Circuit Analysis*, 14th ed. Pearson Education India, 2022

CSE 4107 Structured Programming I Credit: 3.00

Introduction, Programming Concepts, Algorithm and Logic, Constants, Variables, Keywords and Data Types, Operators and expressions, Managing Input and Output Operations, Decision Making and Branching, Decision Making and Looping, Arrays, Multi-dimensional Arrays, Strings, User defined functions, Recursion, Structures and Unions, File Management in C, Pointers, Dynamic Memory Allocation and Linked List, The Preprocessor and some advanced topics, Advanced data types and operators.

Recommended Texts:

1. H. Schildt, *Teach Yourself C*, 3rd ed. McGraw-Hill, 1997

2. E. Balagurusamy, *Programming In Ansi C*, 8th ed. Noida, Uttar Pradesh, India: McGraw Hill Education, 2019

SWE 4101 Introduction to Software Engineering Credit: 3.00

Basic Computer Concepts, Concepts in Hardware and Software.

Basic Computer Organization: Processor and Memory, Secondary Storage Devices, Input-Output Devices, Networking, Introduction to Web and other emerging technologies such as Blogs, Wiki, RSS, Podcasting, Cloud applications.

Computer Software: Programming Languages, Compiler, Assembler, Linker.

Software Engineering: Software Development Life Cycle, Introduction to Software Process Models, Software Requirements Analysis, Software Documentation, Introduction to Software Design, Testing, Deliverables and Maintenance.

Recommended Texts: